📒 AI-BFSI Internship Project Documentation

Scalability & Future Plan – Subscription Waste Detector

1. Introduction

Scalability describes a system's ability to handle larger workloads, datasets, or new requirements without requiring a complete redesign. Future planning defines how the system can evolve while preserving its architecture.

The **Subscription Waste Detector** was developed as a modular Streamlit application with a focused Version 1 scope. While the current implementation targets individual users and local execution, its modular design supports realistic future enhancements such as persistent storage, authentication, cloud deployment, and advanced analytics.

2. Objectives

- Maintain modular architecture
- Support feature expansion
- Improve maintainability
- Enable integration of new technologies
- Prepare for larger datasets and broader usage
- Define a roadmap for future development

3. Current Scalability Assessment

Version 1 provides a foundation for growth through modular design. Current scalability characteristics include:

- Modular application structure
- Independent data processing workflow
- Separate visualization components
- Dedicated AI recommendation module
- Reusable expense analysis pipeline
- Extensible Streamlit interface

4. Current Architecture Strengths

Characteristic	Contribution
Modular Design	Components can be extended independently
Python-Based Processing	Supports additional analytical capabilities
Streamlit Interface	Allows rapid enhancement of UI components
Pandas Data Processing	Efficient handling of structured datasets
Plotly Visualizations	Additional charts can be added without redesign
Groq API Integration	AI capabilities can expand without affecting other modules

5. Current Scalability Limitations

Limitation	Current Status
User Authentication	Not Implemented
Database Storage	Not Implemented
Cloud Deployment	Not Implemented
Multi-user Support	Not Implemented
Bank API Integration	Not Implemented
Email Notifications	Not Implemented
Background Processing	Not Implemented

These exclusions are consistent with the Version 1 scope.

6. Scalability Roadmap

Phase 1 – Version 1 (Completed)

- CSV/Excel upload
- Manual entry
- Subscription detection
- Dashboard visualization
- Renewal reminders
- AI-powered recommendations

Phase 2 – Data Management

- Database integration
- Persistent storage
- Historical expense tracking

Phase 3 – User Management

- Authentication
- Secure login
- Personalized dashboards

Phase 4 – Cloud Deployment

- Cloud hosting
- Online accessibility
- Secure storage

Phase 5 – Advanced Analytics

- Spending trend analysis
- Long-term subscription reports
- Enhanced dashboard metrics

7. Future Enhancement Opportunities

- **User Authentication:** Secure access and personalized dashboards
- **Database Integration:** Persistent storage beyond session data
- **Bank Statement Integration:** Direct import of transaction records
- **Enhanced AI Capabilities:** More detailed financial insights via Groq API
- **Advanced Visualizations:** Long-term spending and category comparisons
- **Notification Services:** Optional reminders via email/SMS

8. Scalability Considerations

Future expansion should preserve modularity:

- Maintain separation of responsibilities
- Keep modules independent
- Reuse existing components
- Extend rather than redesign architecture
- Preserve compatibility with current functionality

9. Benefits of Future Scalability

Enhancement	Expected Benefit
Database Integration	Persistent data storage
User Authentication	Personalized user experience
Cloud Deployment	Improved accessibility
Enhanced Analytics	Better financial insights
Notification Services	Improved subscription awareness
Advanced Reporting	Comprehensive financial analysis

10. Scalability Challenges

- Efficient handling of larger datasets
- Maintaining responsiveness
- Protecting user data and privacy
- Ensuring reliable AI service integration
- Preserving modularity with new features

11. Long-Term Vision

The Subscription Waste Detector aims to evolve from a local AI-assisted financial analysis tool into a comprehensive subscription management platform. Future versions will emphasize simplicity, usability, and practical insights, ensuring new features directly support the core purpose of managing recurring subscription expenses.

12. Conclusion

Version 1 of the Subscription Waste Detector provides a scalable foundation through modular architecture and independent components. While the current scope is intentionally limited, the design supports future enhancements without major redesign.